IPL Data Analysis Assignment

```
import pandas as pd
import matplotlib.pyplot as plt
```

Load Dataset

```
matches = pd.read_csv('matches.csv')
deliveries = pd.read_csv('deliveries.csv')
matches.head()
```

```
   id  season       city        date                          team1  \
0   1    2017  Hyderabad  2017-04-05          Sunrisers Hyderabad
1   2    2017       Pune  2017-04-06               Mumbai Indians
2   3    2017     Rajkot  2017-04-07                Gujarat Lions
3   4    2017     Indore  2017-04-08       Rising Pune Supergiant
4   5    2017  Bangalore  2017-04-08  Royal Challengers Bangalore

                          team2                  toss_winner
toss_decision  \
0  Royal Challengers Bangalore  Royal Challengers Bangalore
field
1       Rising Pune Supergiant       Rising Pune Supergiant
field
2        Kolkata Knight Riders        Kolkata Knight Riders
field
3              Kings XI Punjab              Kings XI Punjab
field
4             Delhi Daredevils  Royal Challengers Bangalore
bat

   result  dl_applied                       winner  win_by_runs  \
0  normal           0          Sunrisers Hyderabad           35
1  normal           0       Rising Pune Supergiant            0
2  normal           0        Kolkata Knight Riders            0
3  normal           0              Kings XI Punjab            0
4  normal           0  Royal Challengers Bangalore           15

   win_by_wickets player_of_match
venue  \
0               0    Yuvraj Singh  Rajiv Gandhi International Stadium,
Uppal
1               7       SPD Smith    Maharashtra Cricket Association
Stadium
2              10         CA Lynn     Saurashtra Cricket Association
Stadium
3               6      GJ Maxwell                     Holkar Cricket
Stadium
```

```
4                0       KM Jadhav                        M Chinnaswamy
Stadium

          umpire1        umpire2 umpire3
0     AY Dandekar       NJ Llong     NaN
1  A Nand Kishore         S Ravi     NaN
2     Nitin Menon      CK Nandan     NaN
3    AK Chaudhary  C Shamshuddin     NaN
4             NaN            NaN     NaN
```

1. Match won by maximum runs

```
max_run = matches.loc[matches['win_by_runs'].idxmax()]
print(max_run[['season','team1','team2','winner','win_by_runs']])
```

```
season                      2017
team1            Mumbai Indians
team2          Delhi Daredevils
winner           Mumbai Indians
win_by_runs                 146
Name: 43, dtype: object
```

2. Match won by maximum wickets

```
max_wicket = matches.loc[matches['win_by_wickets'].idxmax()]
print(max_wicket[['season','team1','team2','winner','win_by_wickets']]
)
```

```
season                             2017
team1                    Gujarat Lions
team2             Kolkata Knight Riders
winner            Kolkata Knight Riders
win_by_wickets                       10
Name: 2, dtype: object
```

3. Match won by minimum runs

```
min_run_match = matches[matches['win_by_runs'] > 0]
min_run = min_run_match.loc[min_run_match['win_by_runs'].idxmin()]

print(min_run[['season','team1','team2','winner','win_by_runs']])
```

```
season                            2017
team1                  Mumbai Indians
team2          Rising Pune Supergiant
winner                 Mumbai Indians
win_by_runs                         1
Name: 58, dtype: object
```

4. Match won by minimum wickets

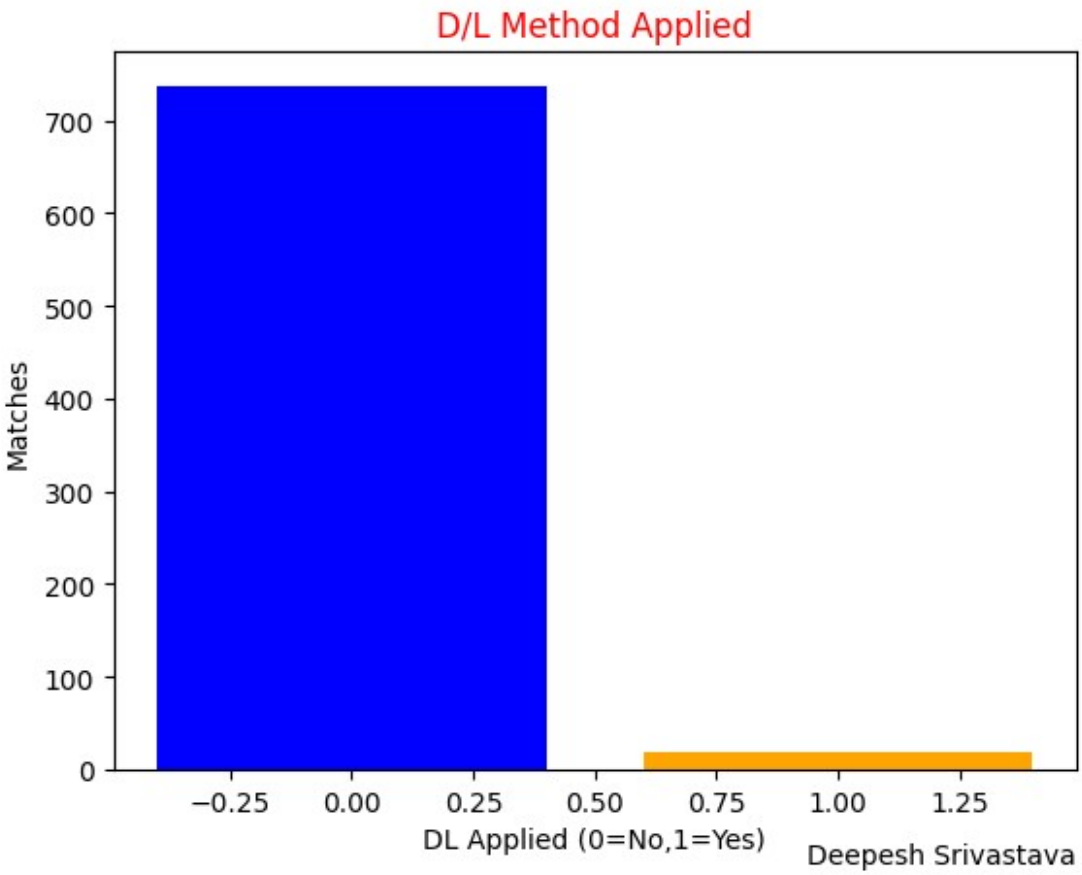
```
min_wicket_match = matches[matches['win_by_wickets'] > 0]
min_wicket =
min_wicket_match.loc[min_wicket_match['win_by_wickets'].idxmin()]

print(min_wicket[['season','team1','team2','winner','win_by_wickets']]
)
```

```
season                                2015
team1                   Kings XI Punjab
team2             Kolkata Knight Riders
winner            Kolkata Knight Riders
win_by_wickets                        1
Name: 559, dtype: object
```

5. Matches where D/L applied vs not applied

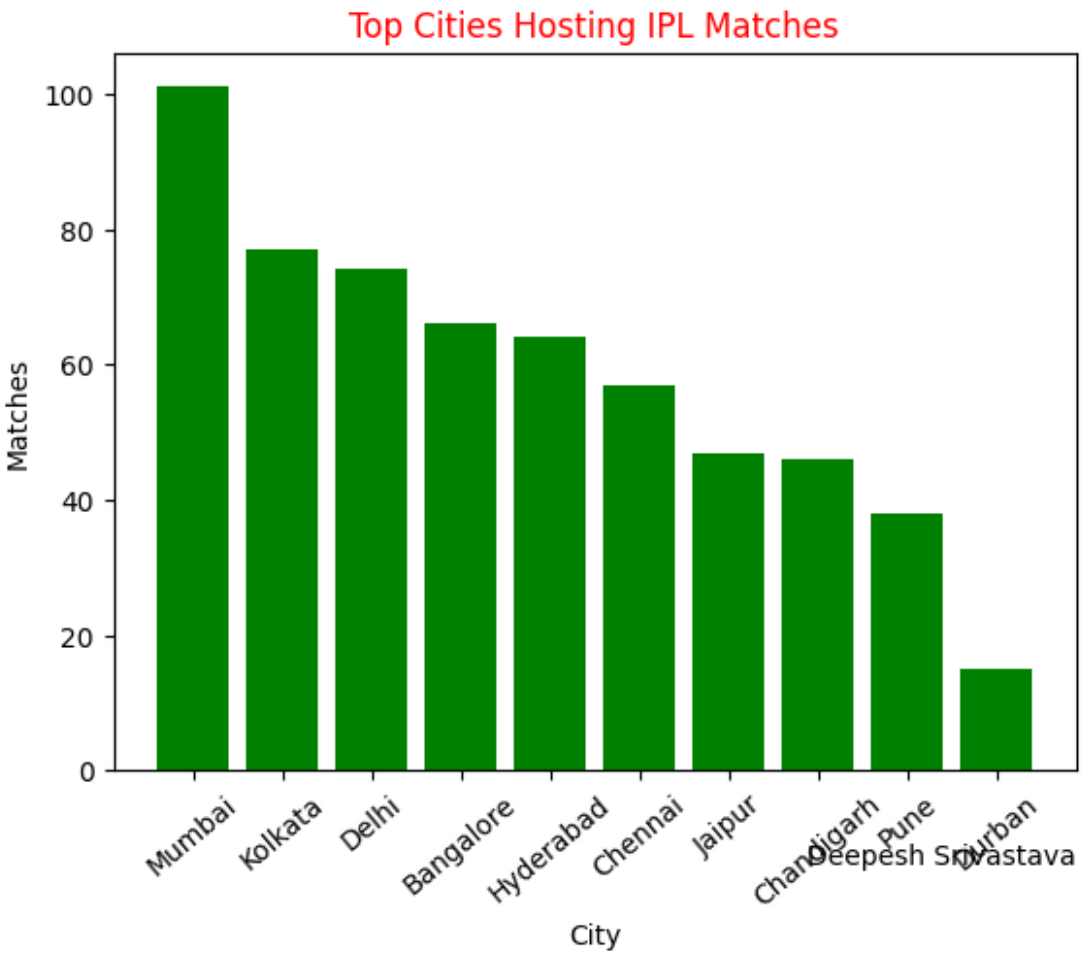
```
dl = matches['dl_applied'].value_counts()
plt.bar(dl.index, dl.values , color=['blue', 'orange'])
plt.title("D/L Method Applied",color='red')
plt.xlabel("DL Applied (0=No,1=Yes)")
plt.ylabel("Matches")
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS Most IPL matches are completed without using the D/L method. This shows that rain affects only a small number of matches.

6. Number of matches in each city

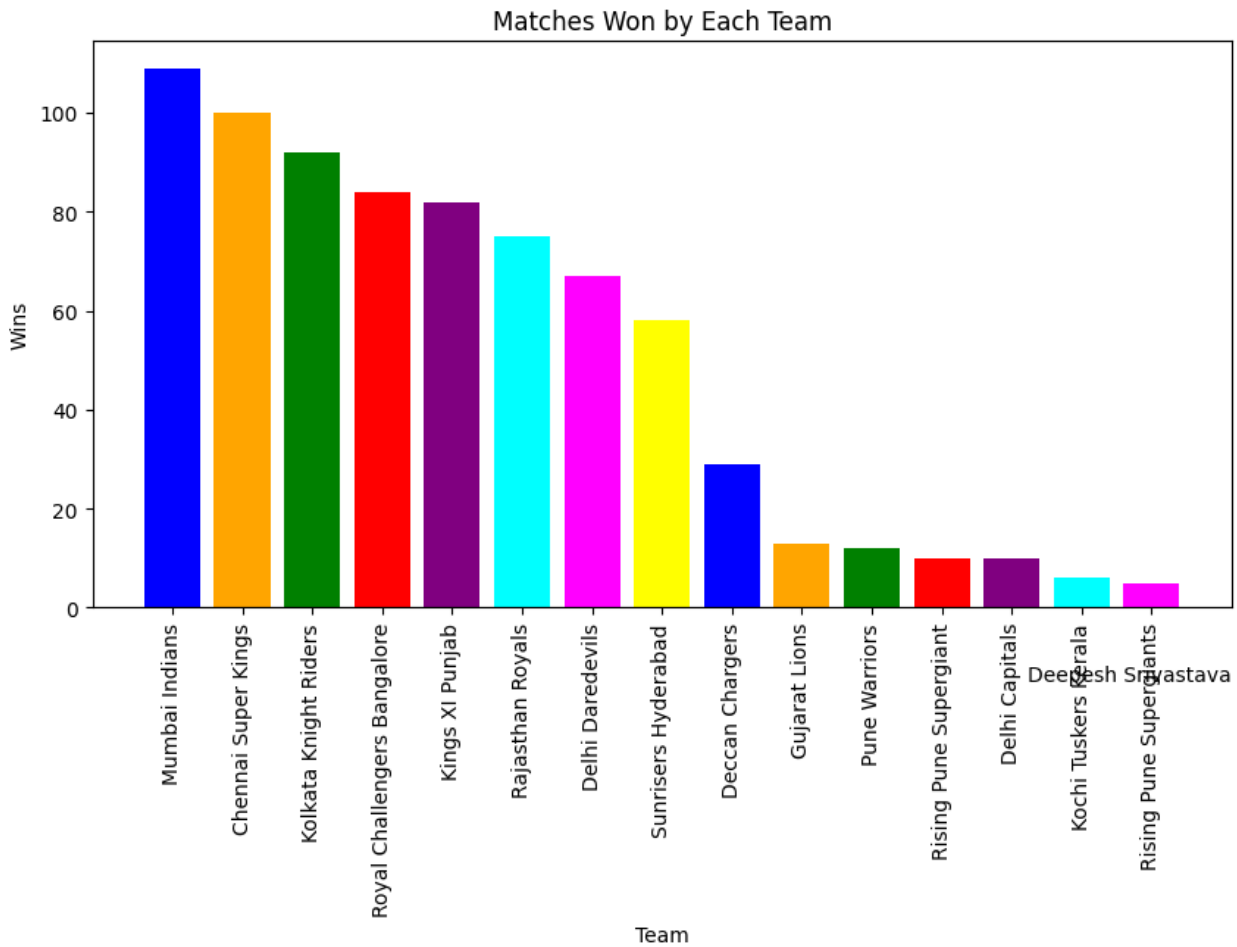
```
city = matches['city'].value_counts().head(10)
plt.bar(city.index, city.values, color='green')
plt.xticks(rotation=40)
plt.title("Top Cities Hosting IPL Matches", color='red')
plt.xlabel("City")
plt.ylabel("Matches")
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS Mainly, most of the matches are held in Mumbai and Kolkata.

7. Matches won by each team

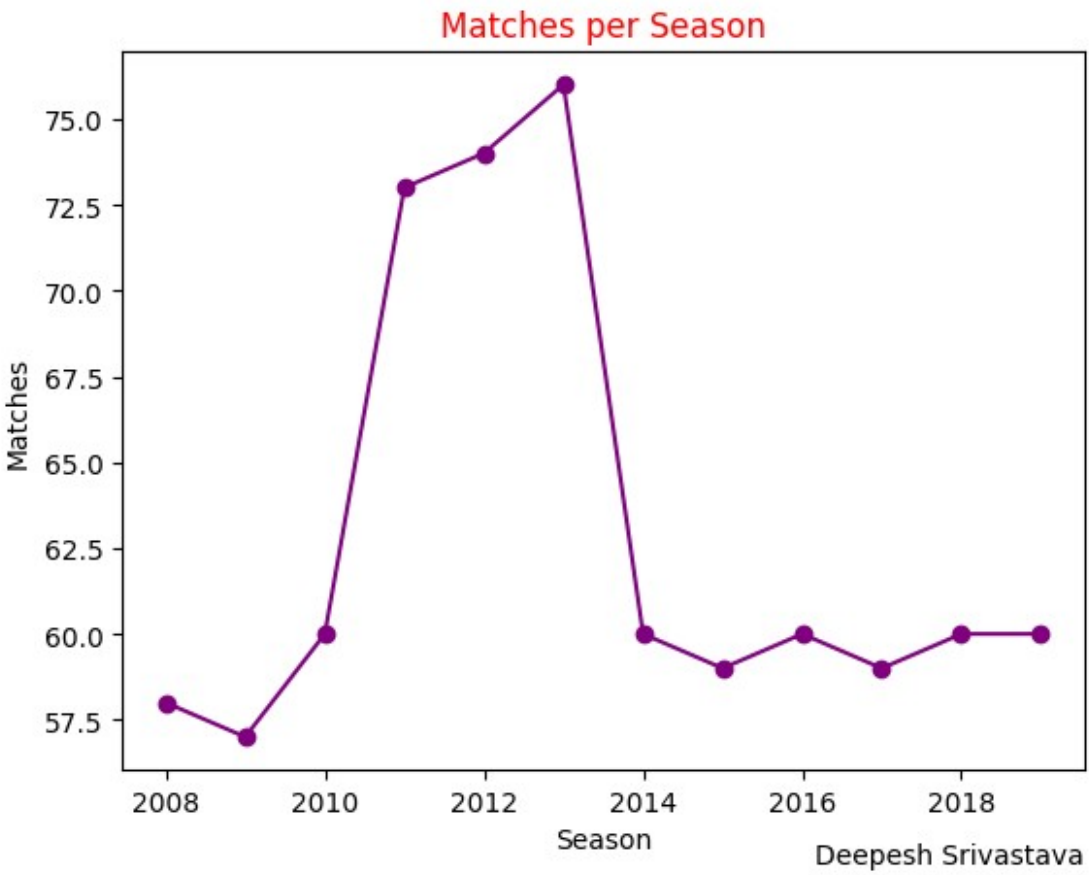
```
team_wins = matches['winner'].value_counts()
plt.figure(figsize=(10,5))
plt.bar(team_wins.index, team_wins.values, color=['blue', 'orange',
'green', 'red', 'purple', 'cyan', 'magenta', 'yellow'])
plt.xticks(rotation=90)
plt.title("Matches Won by Each Team")
plt.xlabel("Team")
plt.ylabel("Wins")
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS Mumbai Indians have won the most matches in the IPL. Top 2nd highest team is Chennai Super Kings 3ed position team is kolkata Knights Riders

8. Matches held every season

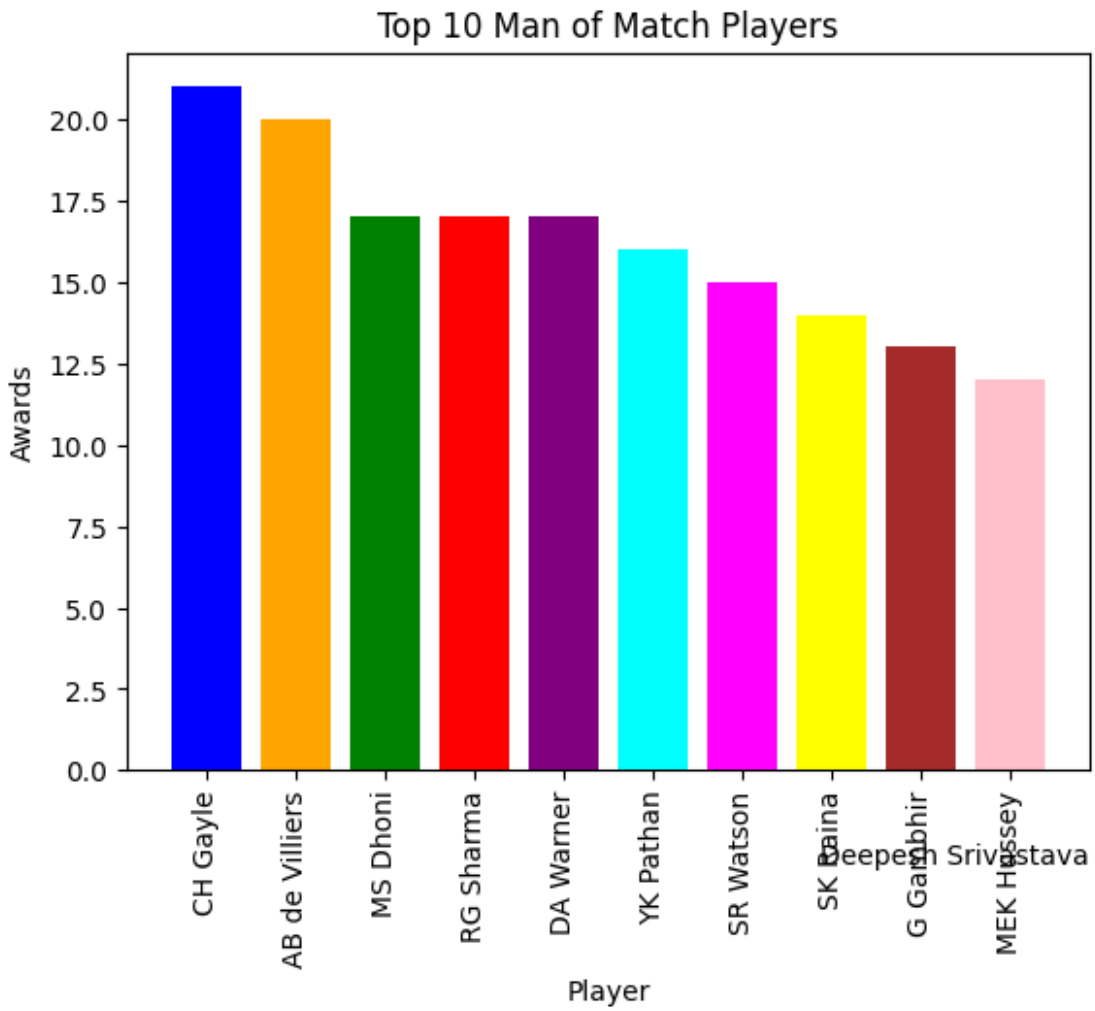
```
season = matches['season'].value_counts().sort_index()
plt.plot(season.index, season.values, marker='o', color='purple')
plt.title("Matches per Season", color='red')
plt.xlabel("Season")
plt.ylabel("Matches")
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS The highest number of matches were played in the years 2012-2014.

9. Top 10 players with most Man of the Match awards

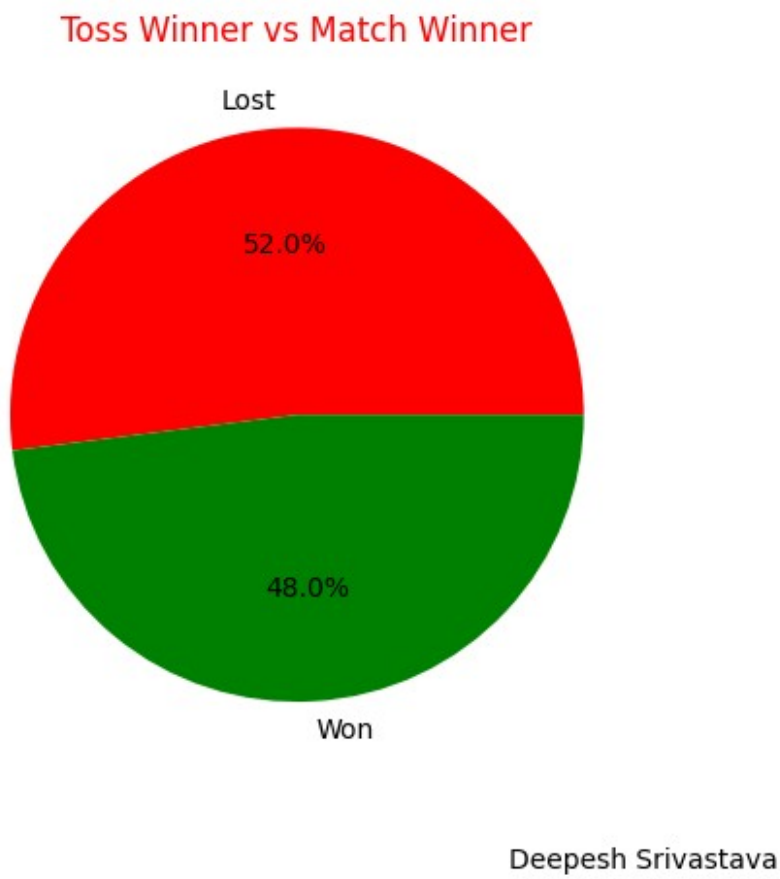
```
mom = matches['player_of_match'].value_counts().head(10)
plt.bar(mom.index, mom.values, color=['blue', 'orange', 'green',
'red', 'purple', 'cyan', 'magenta', 'yellow', 'brown', 'pink'])
plt.xticks(rotation=90)
plt.title("Top 10 Man of Match Players")
plt.xlabel("Player")
plt.ylabel("Awards")
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS CH Gayle is the 1st position of man of the matches in IPL Season. 2nd most highest man of the matches in IPL Season is AB de Villers

10. Does winning toss mean winning match

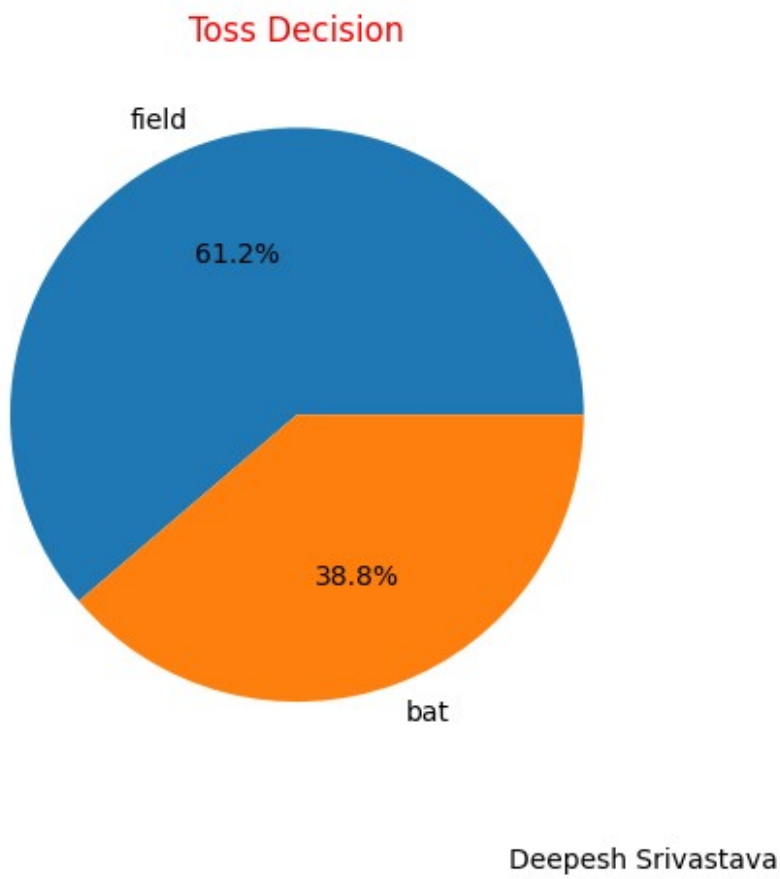
```
matches['toss_match_win'] = matches['toss_winner'] ==
matches['winner']
result = matches['toss_match_win'].value_counts()
plt.pie(result.values, labels=['Lost','Won'], autopct='%1.1f%
%',colors=['red','green'])
plt.title("Toss Winner vs Match Winner" ,color='red')
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS The result mainly depends on how well the team plays. Toss gives a small advantage, but performance is more important.

11. Toss decision taken by captains

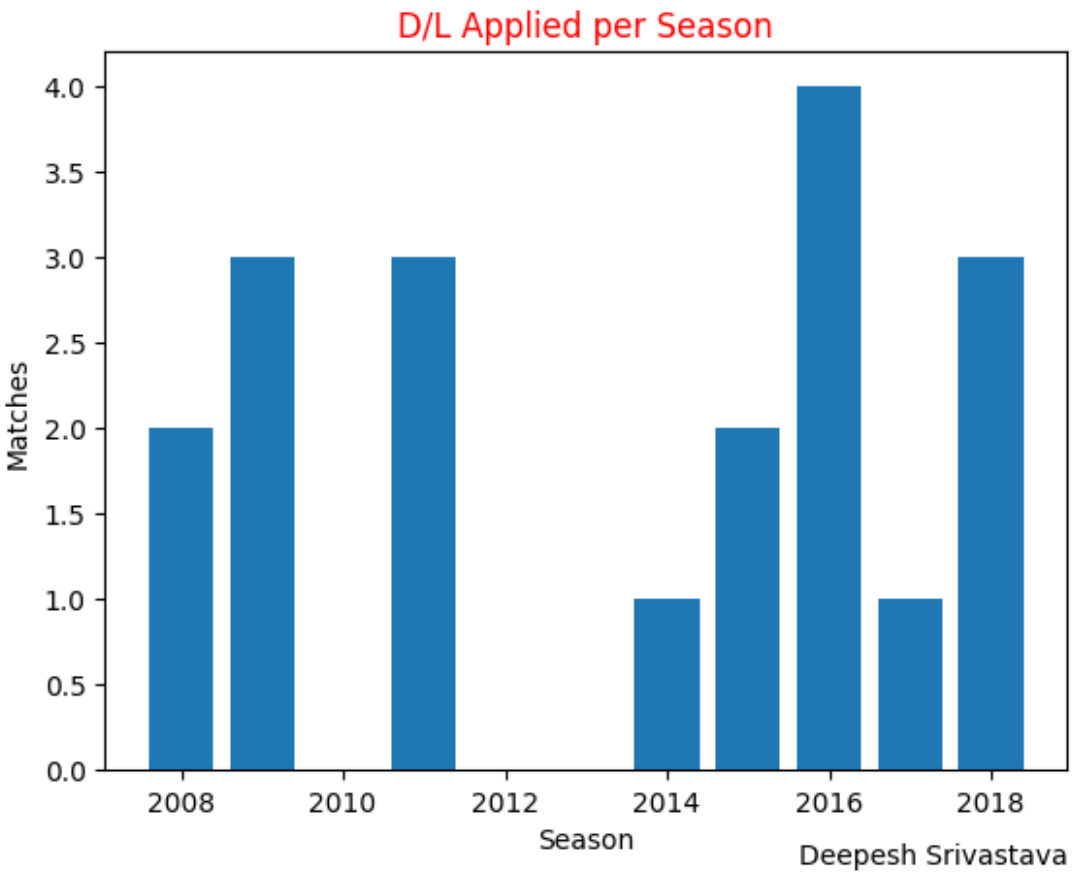
```
decision = matches['toss_decision'].value_counts()
plt.pie(decision.values, labels=decision.index, autopct='%1.1f%%')
plt.title("Toss Decision",color='red')
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS Most captains prefer to field first after winning the toss. Chasing a target is often considered easier in T20 matches.

12. D/L method applied every season

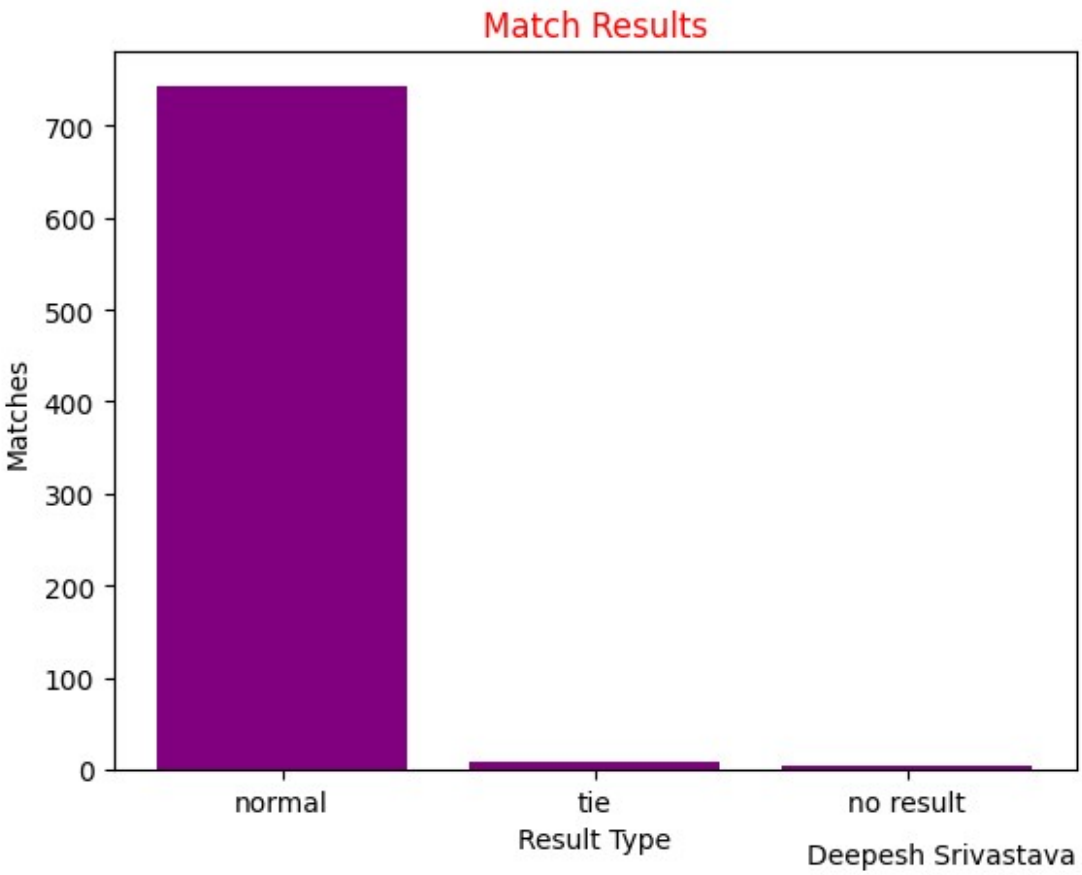
```
dl_season = matches[matches['dl_applied']==1]
['season'].value_counts().sort_index()
plt.bar(dl_season.index, dl_season.values)
plt.title("D/L Applied per Season" ,color='red')
plt.xlabel("Season")
plt.ylabel("Matches")
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS Only a few seasons had matches affected by rain. The D/L method helps adjust the target when rain interrupts the game.

13. Different results for matches

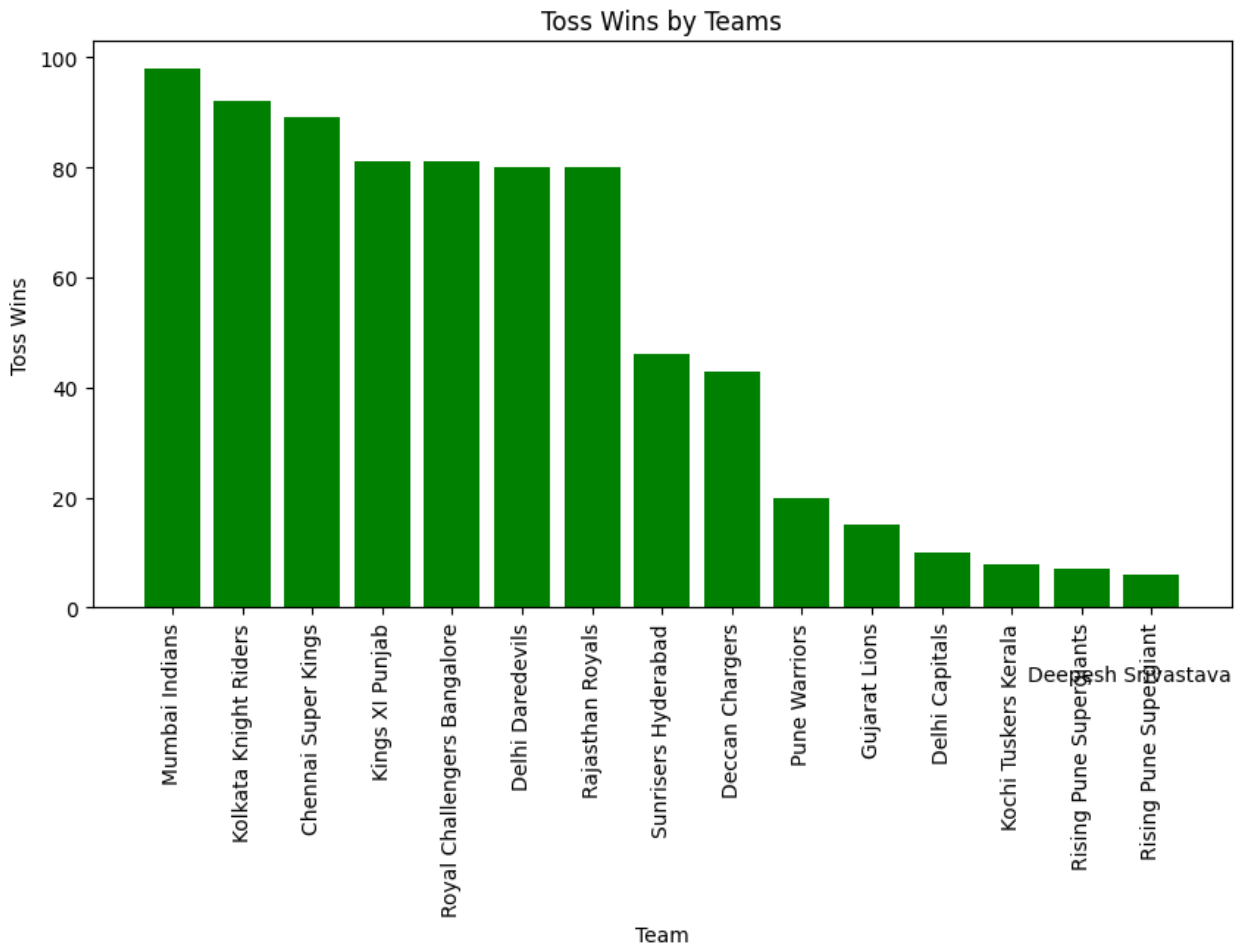
```
results = matches['result'].value_counts()
plt.bar(results.index, results.values, color='purple')
plt.title("Match Results" ,color='red')
plt.xlabel("Result Type")
plt.ylabel("Matches")
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS Most matches end with a normal result where one team wins. Some matches end in a tie or no result. In tied matches, a Super Over is sometimes used to decide the winner.

14. Number of toss wins by each team

```
toss_wins = matches['toss_winner'].value_counts()
plt.figure(figsize=(10,5))
plt.bar(toss_wins.index, toss_wins.values, color='green')
plt.xticks(rotation=90)
plt.title("Toss Wins by Teams")
plt.xlabel("Team")
plt.ylabel("Toss Wins")
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS The Highest number of toss winnig team is Mumbai Indians The toss helps teams choose their match strategy.

15. Best venue for defending and chasing

```
defend = matches[matches['win_by_runs']>0]
['venue'].value_counts().head(5)
chase = matches[matches['win_by_wickets']>0]
['venue'].value_counts().head(5)
print(" Best venues for defending")
print(defend)
print("Best venues for chasing")
print(chase)
```

```
 Best venues for defending
venue
Wankhede Stadium                    35
Feroz Shah Kotla                    33
Eden Gardens                        32
M Chinnaswamy Stadium               30
MA Chidambaram Stadium, Chepauk     30
```

```
Name: count, dtype: int64
Best venues for chasing
venue
Eden Gardens                                    45
M Chinnaswamy Stadium                           40
Wankhede Stadium                                37
Feroz Shah Kotla                                33
Rajiv Gandhi International Stadium, Uppal       32
Name: count, dtype: int64
```

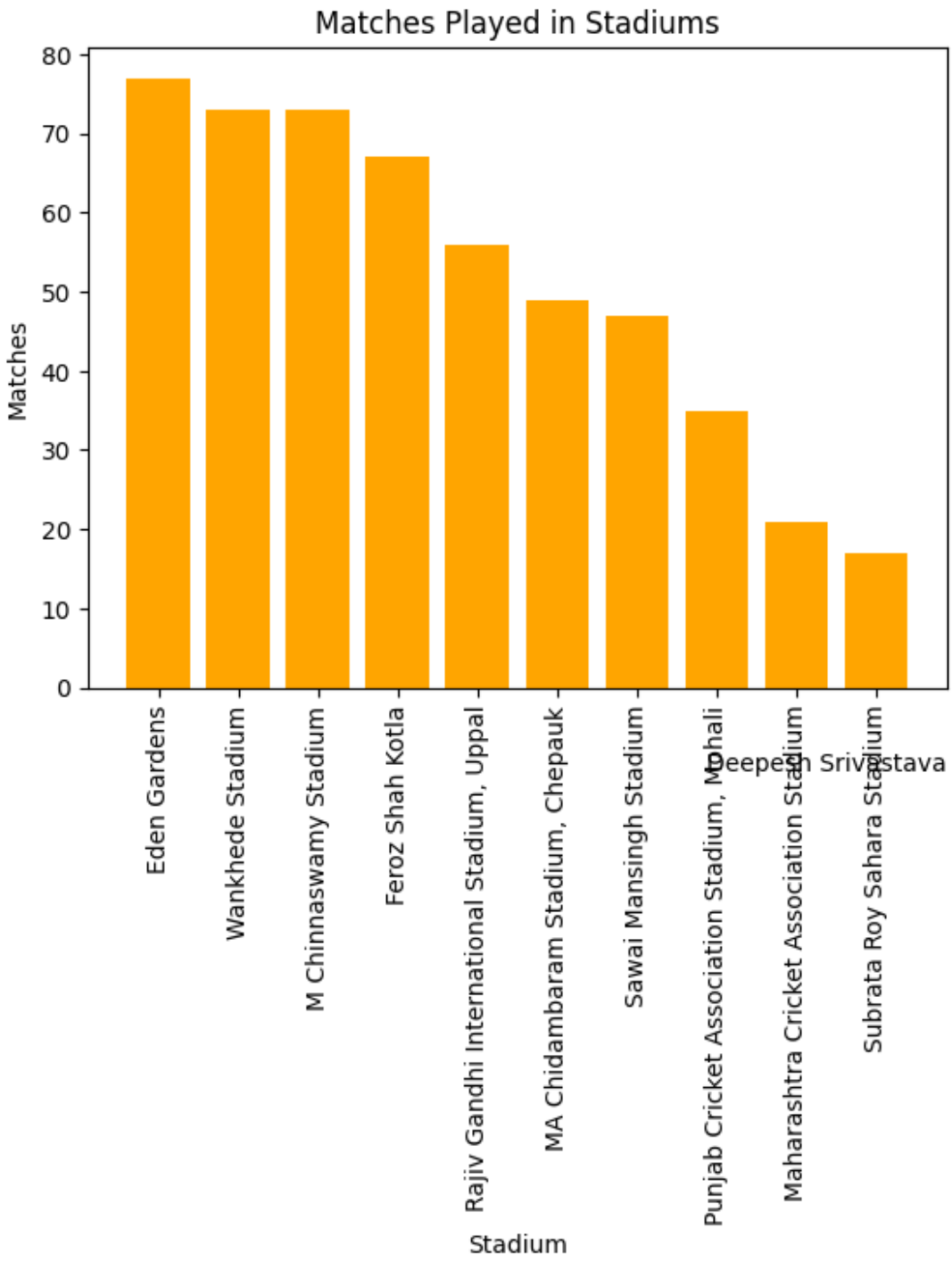
16. Best defending and chasing teams

```
defending = matches[matches['win_by_runs']>0]
['winner'].value_counts().head(5)
chasing = matches[matches['win_by_wickets']>0]
['winner'].value_counts().head(5)
print("Best Defending Teams")
print(defending)
print("Best Chasing Teams")
print(chasing)
```

```
Best Defending Teams
winner
Mumbai Indians                  57
Chennai Super Kings             52
Kings XI Punjab                 38
Kolkata Knight Riders           36
Royal Challengers Bangalore     35
Name: count, dtype: int64
Best Chasing Teams
winner
Kolkata Knight Riders           56
Mumbai Indians                  50
Royal Challengers Bangalore     48
Chennai Super Kings             48
Rajasthan Royals                46
Name: count, dtype: int64
```

17. Matches played in different stadiums

```
venue = matches['venue'].value_counts().head(10)
plt.bar(venue.index, venue.values, color='orange')
plt.xticks(rotation=90)
plt.title("Matches Played in Stadiums")
plt.xlabel("Stadium")
plt.ylabel("Matches")
plt.figtext(0.9, 0.01, "Deepesh Srivastava",
horizontalalignment='right')
plt.show()
```



INSIGHTS Some stadiums host more matches than others. These stadiums are usually the home grounds of IPL teams. They also have large capacity and good facilities for matches.